DSP_POWER

File: DSP_POWER.kicad_sch

DSP-I

File: DSP-I.kicad_sch

DSP-II

File: DSP-II.kicad_sch

Serial_SATA

File: Serial_SATA.kicad_sch

SD_Card_LCD

File: SD_Card_LCD.kicad_sch

Video_Decoder (page 12)

File: Video_Decoder.kicad_sch

Power_Supply-I

File: Power_Supply-I.kicad_sch

Power_Supply_II

File: Power_Supply_II.kicad_sch

LCD_II

File: LCD_II.kicad_sch

DDR_RAM

File: DDR_RAM.kicad_sch

NAND_Flash

File: NAND_Flash.kicad_sch

Sound_Coded

File: Sound_Coded.kicad_sch

LAN_8710A

File: LAN_8710A.kicad_sch

Connectors

File: Connectors.kicad_sch

Mislleneous

File: Mislleneous.kicad_sch

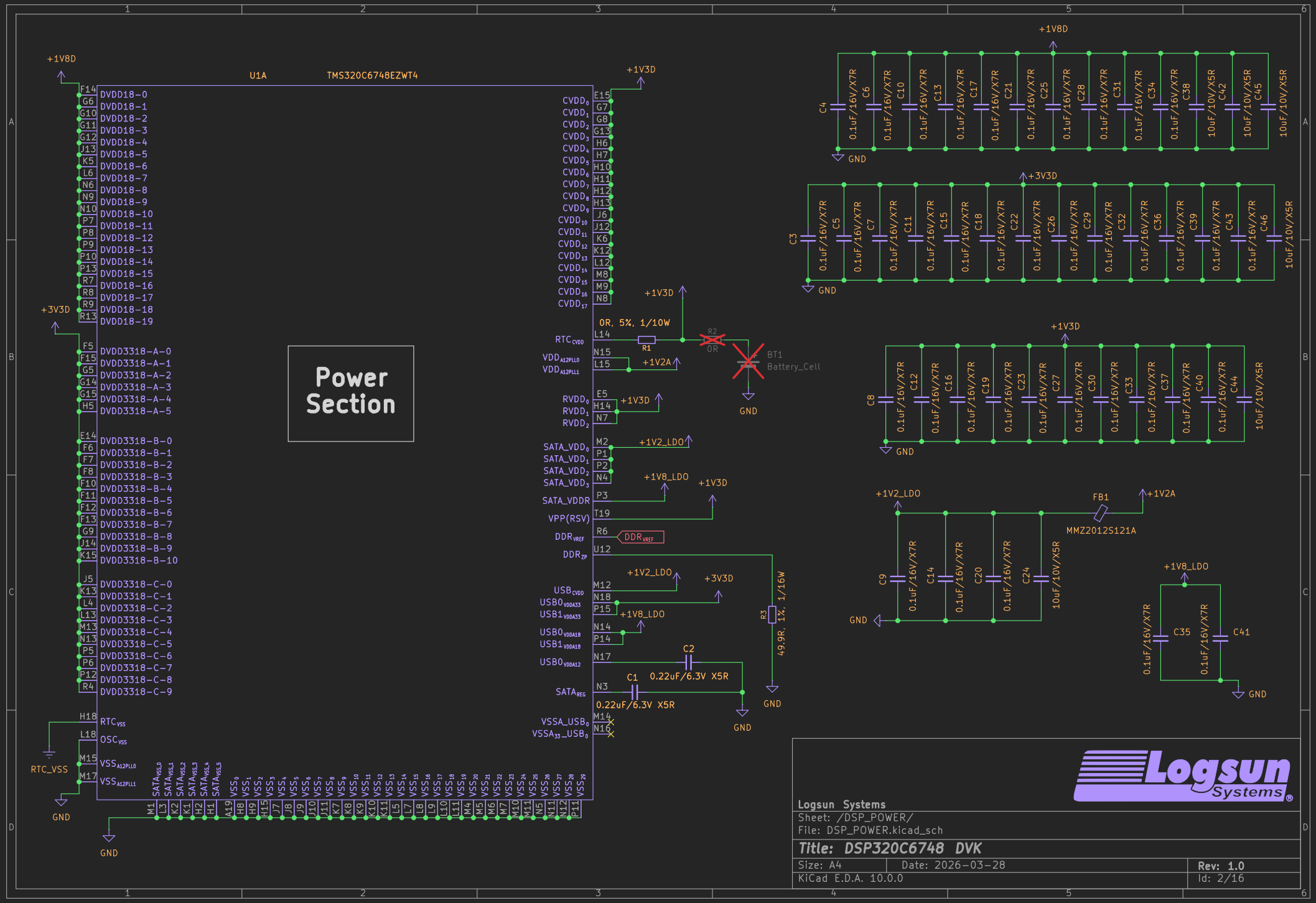
Logsun Systems

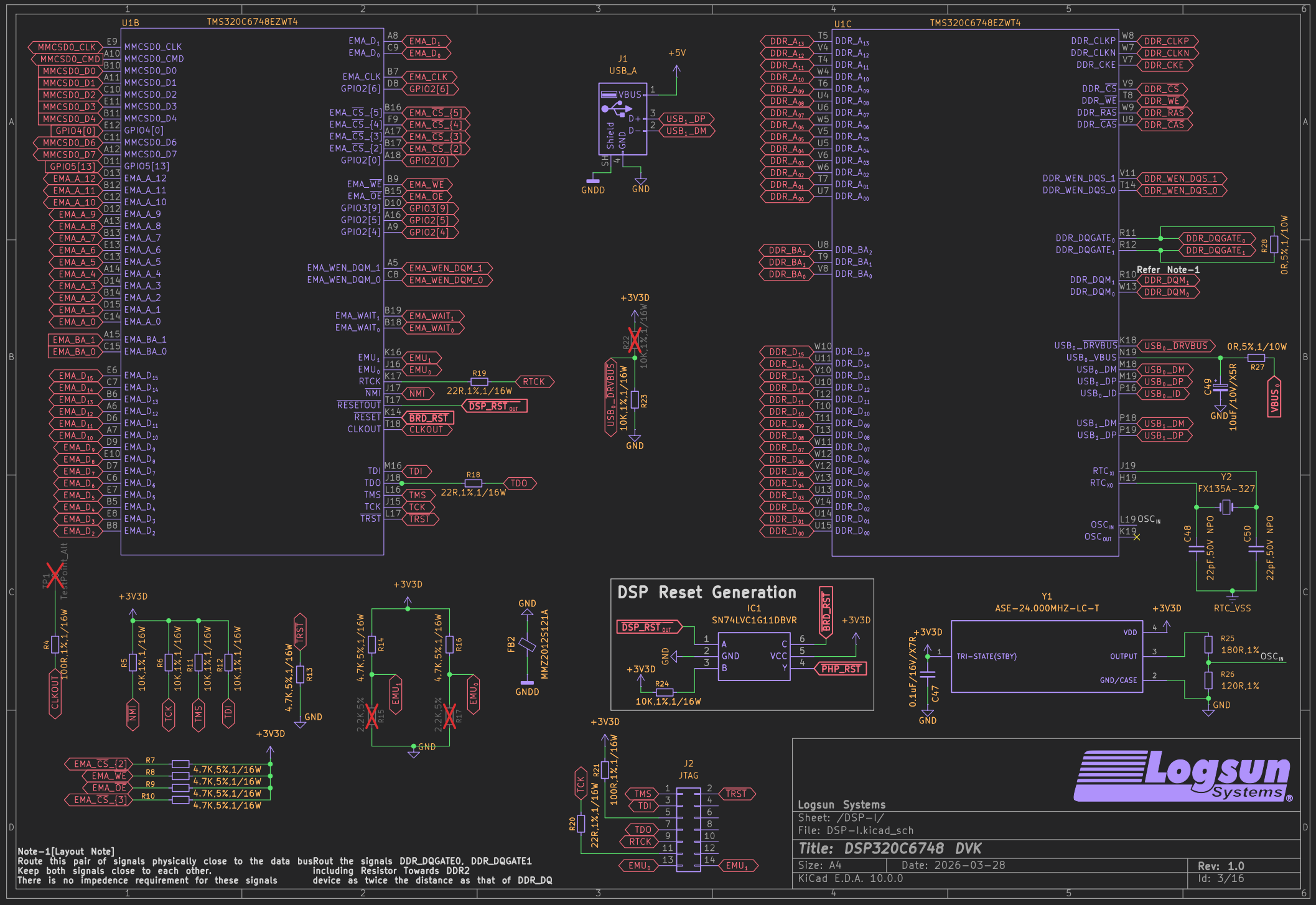
Sheet: /

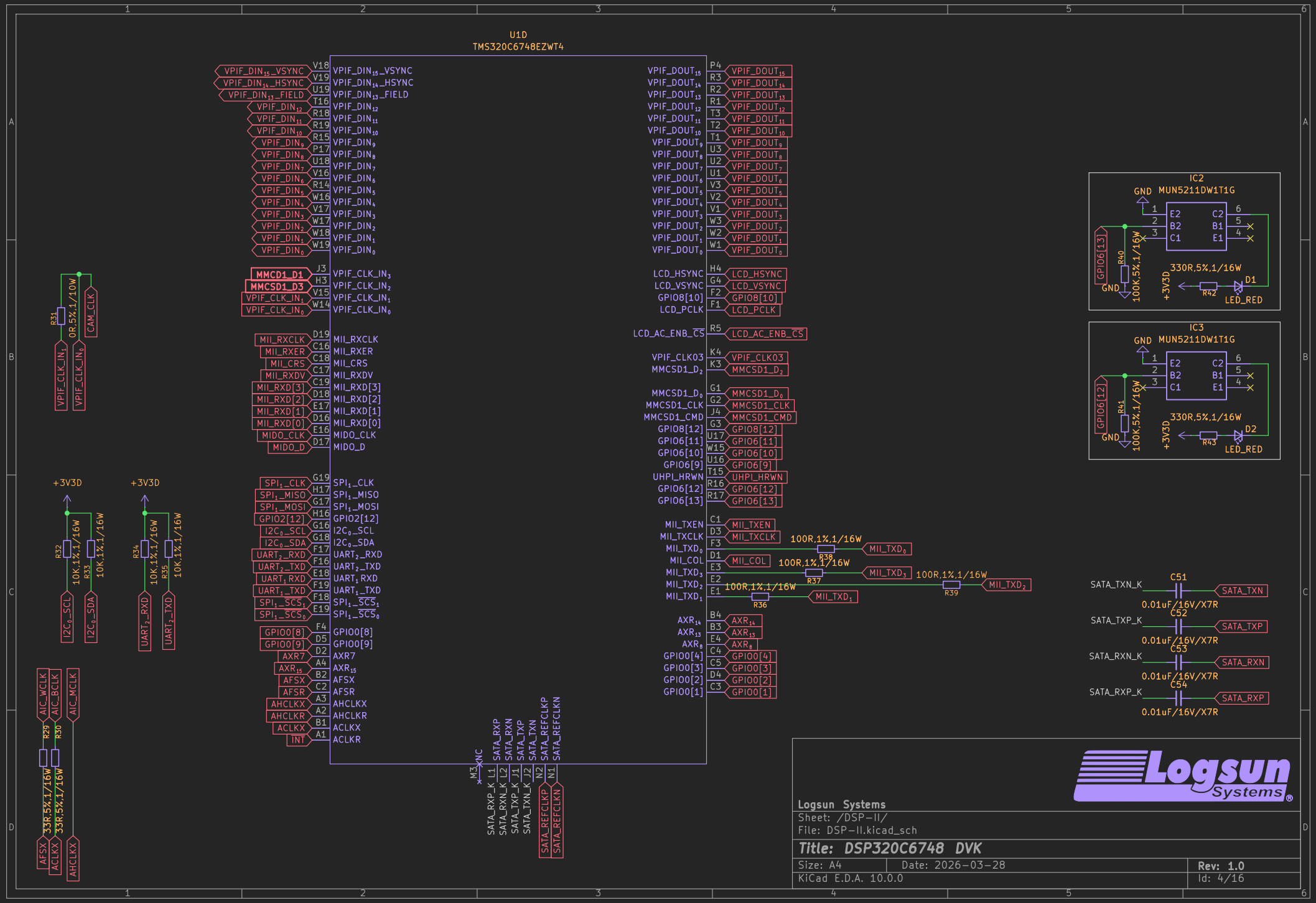
File: DSP320C6748_DVK.kicad_sch

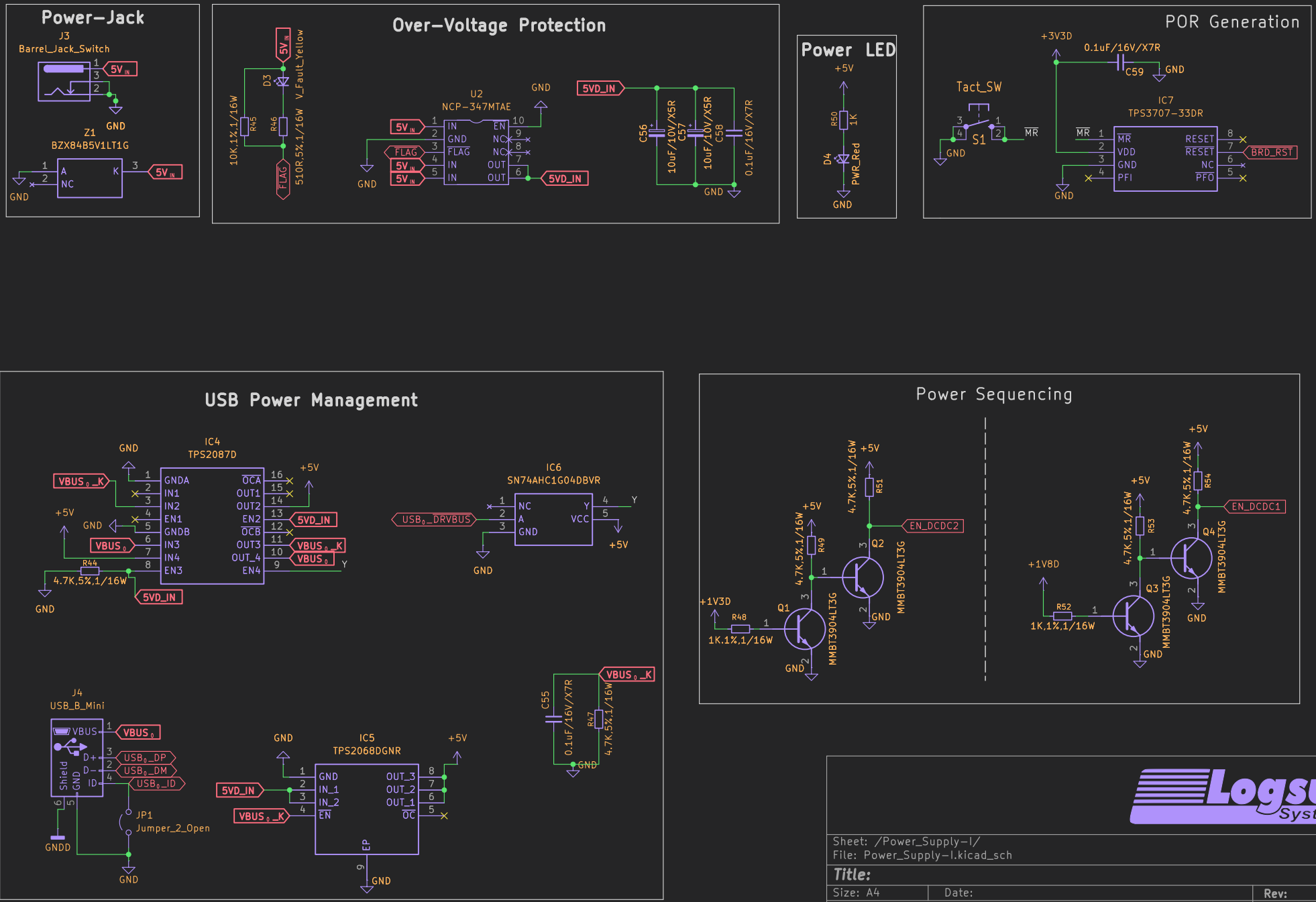
Title: DSP320C6748 DVK

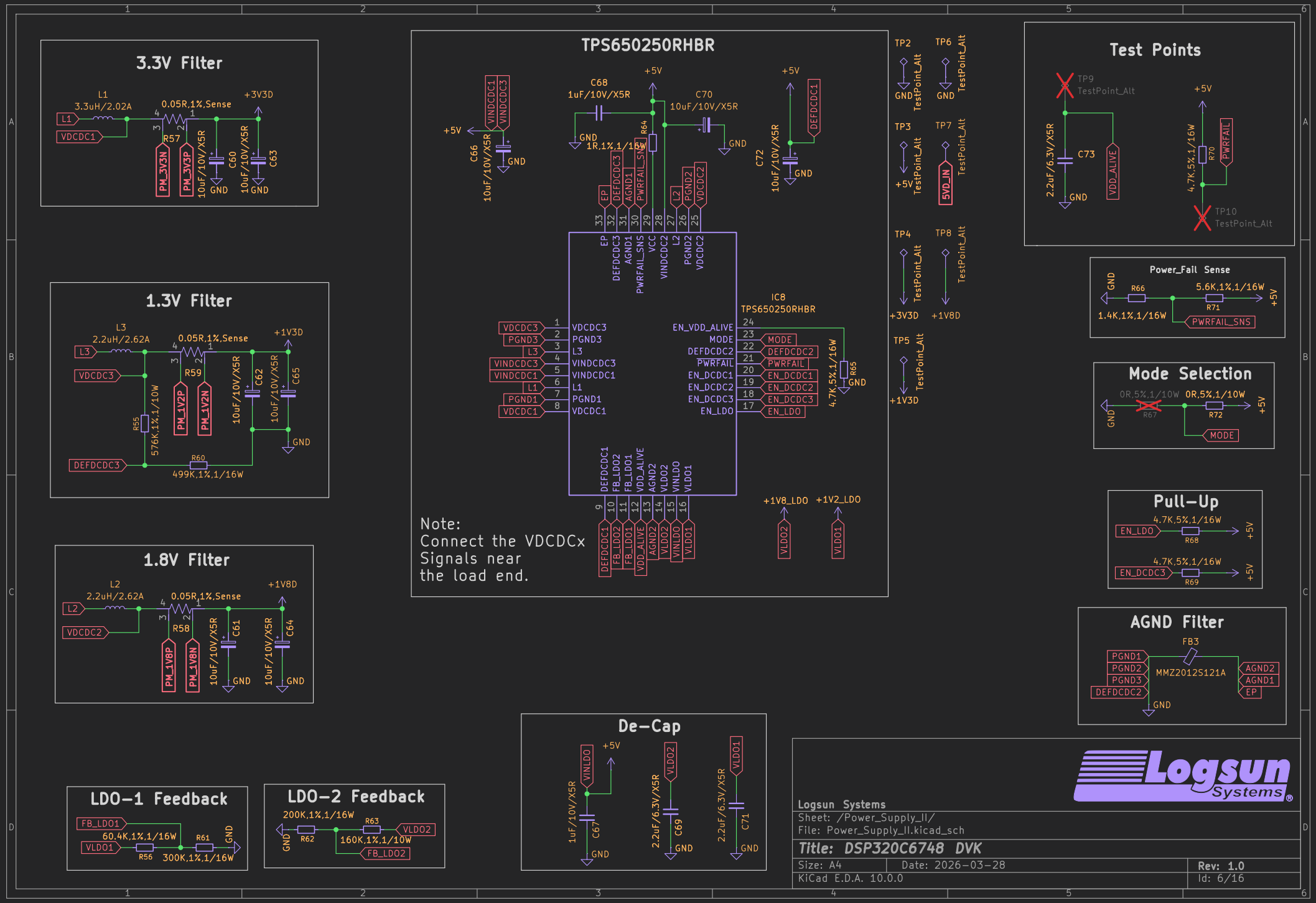
Size: A4	Date: 2026-03-28	Rev: 1.0
KiCad E.D.A. 10.0.0		Id: 1/16

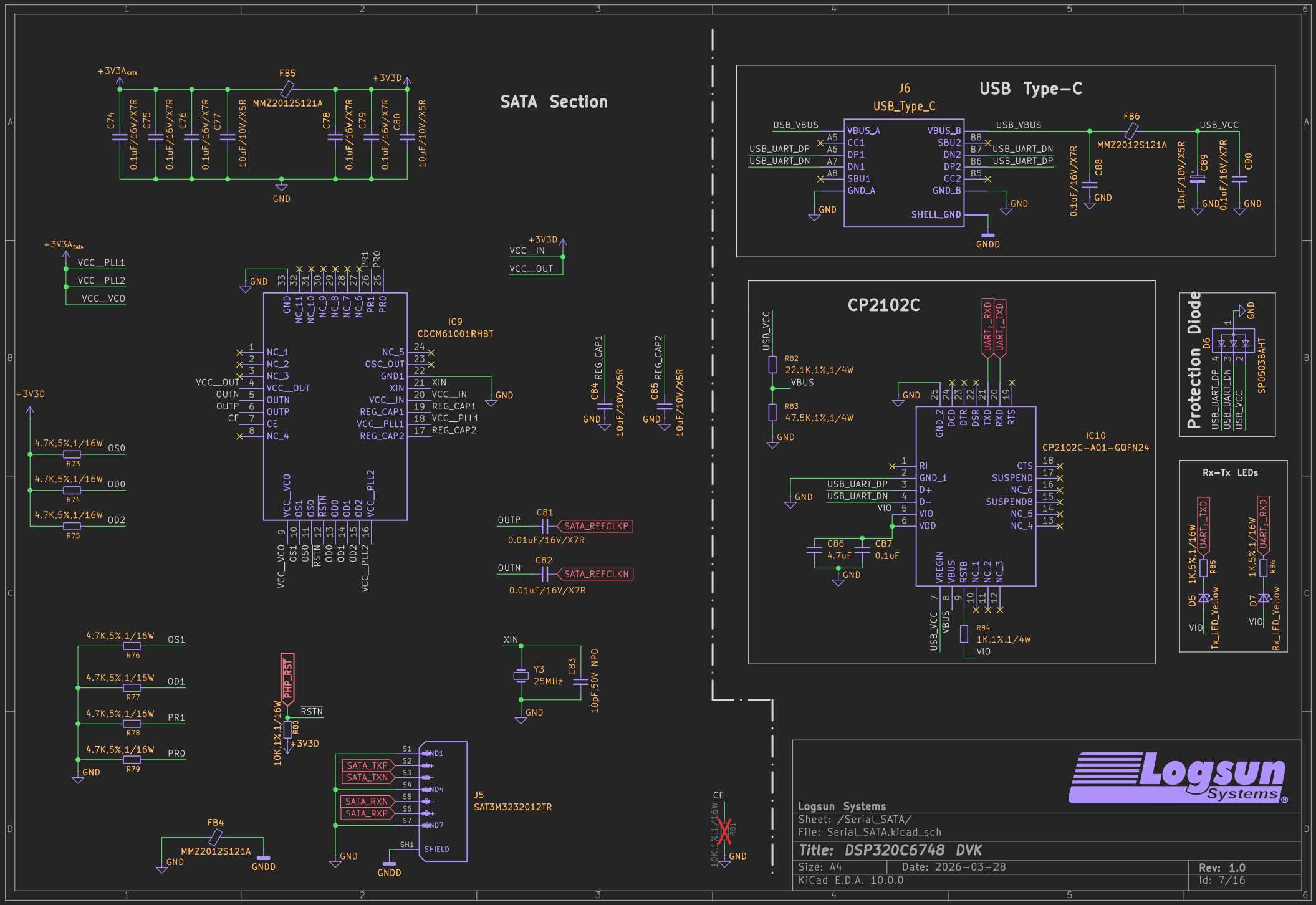

U1A
TMS320C6748EZWT4
Power Section
+1V8D
+3V3D
+1V3D
+1V2A
+1V2_LDO
+1V8_LDO
GND
RTC_VSS
OR, 5%, 1/10W
R1
R2
OR
BT1
Battery_Cell
DDR_VREF
0.22uF/6.3V X5R
C1
C2
R3
49.9R, 1%, 1/16W
FB1
MMZ2012S121A
Logsun Systems
Sheet: /DSP_POWER/
File: DSP_POWER.kicad_sch
Title: DSP320C6748 DVK
Size: A4
Date: 2026-03-28
Rev: 1.0
KiCad E.D.A. 10.0.0
Id: 2/16


U1B TMS320C6748EZWT4
U1C TMS320C6748EZWT4
J1 USB_A
J2 JTAG
IC1 SN74LVC1G11DBVR
Y1 ASE-24.000MHZ-LC-T
Y2 FX135A-327
DSP Reset Generation
Logsun Systems
Sheet: /DSP-I/
File: DSP-I.kicad_sch
Title: DSP320C6748 DVK
Size: A4
Date: 2026-03-28
Rev: 1.0
KiCad E.D.A. 10.0.0
Id: 3/16
Note-1[Layout Note]
Route this pair of signals physically close to the data bus
Keep both signals close to each other.
There is no impedence requirement for these signals
Rout the signals DDR_DQGATE0, DDR_DQGATE1 including Resistor Towards DDR2 device as twice the distance as that of DDR_DQ


U1D
TMS320C6748EZWT4
IC2
MUN5211DW1T1G
IC3
MUN5211DW1T1G
330R,5%,1/16W
LED_RED
100K,5%,1/16W
+3V3D
GND
10K,1%,1/16W
33R,5%,1/16W
0R,5%,1/10W
100R,1%,1/16W
0.01uF/16V/X7R
SATA_TXN_K
SATA_TXP_K
SATA_RXN_K
SATA_RXP_K
Logsun Systems
Sheet: /DSP-II/
File: DSP-II.kicad_sch
Title: DSP320C6748 DVK
Size: A4
Date: 2026-03-28
Rev: 1.0
KiCad E.D.A. 10.0.0
Id: 4/16


Power-Jack
J3
Barrel_Jack_Switch
5V_IN
GND
Z1
BZX84B5V1LT1G
A
NC
K
Over-Voltage Protection
D3
V_Fault_Yellow
10K,1%,1/16W
R45
R46
510R,5%,1/16W
FLAG
U2
NCP-347MTAE
IN
GND
FLAG
EN
NC
OUT
5VD_IN
C56
10uF/10V/X5R
C57
C58
0.1uF/16V/X7R
Power LED
+5V
R50
1K
D4
PWR_Red
POR Generation
+3V3D
0.1uF/16V/X7R
C59
Tact_SW
S1
MR
IC7
TPS3707-33DR
VDD
PFI
RESET
PFO
BRD_RST
USB Power Management
IC4
TPS2087D
GNDA
IN1
IN2
EN1
GNDB
IN3
IN4
EN3
OCA
OUT1
OUT2
EN2
OCB
OUT3
OUT_4
EN4
VBUS0_K
VBUS0
R44
4.7K,5%,1/16W
IC6
SN74AHC1G04DBVR
NC
A
VCC
Y
USB0_DRVBUS
J4
USB_B_Mini
VBUS
D+
D-
ID
Shield
GND
USB0_DP
USB0_DM
USB0_ID
JP1
Jumper_2_Open
GNDD
IC5
TPS2068DGNR
IN_1
IN_2
EN
OUT_3
OUT_2
OUT_1
OC
EP
C55
0.1uF/16V/X7R
R47
4.7K,5%,1/16W
Power Sequencing
R48
1K.1%,1/16W
Q1
MMBT3904LT3G
R49
4.7K,5%,1/16W
R51
Q2
EN_DCDC2
+1V8D
R52
1K,1%,1/16W
R53
Q3
R54
Q4
EN_DCDC1
Logsun Systems
Sheet: /Power_Supply-1/
File: Power_Supply-1.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.0
Id: 5/16


3.3V Filter
L1
3.3uH/2.02A
0.05R,1%,Sense
+3V3D
VDCDC1
R57
PM_3V3N
PM_3V3P
C60
10uF/10V/X5R
C63
10uF/10V/X5R
GND
1.3V Filter
L3
2.2uH/2.62A
0.05R,1%,Sense
+1V3D
VDCDC3
R59
PM_1V2P
PM_1V2N
C62
10uF/10V/X5R
C65
10uF/10V/X5R
R55
576K,1%,1/10W
DEFDCDC3
R60
499K,1%,1/16W
GND
1.8V Filter
L2
2.2uH/2.62A
0.05R,1%,Sense
+1V8D
VDCDC2
R58
PM_1V8P
PM_1V8N
C61
10uF/10V/X5R
C64
10uF/10V/X5R
GND
TPS650250RHBR
+5V
VINDCDC1
VINDCDC3
C66
10uF/10V/X5R
GND
C68
1uF/10V/X5R
C70
10uF/10V/X5R
R64
1R,1%,1/16W
C72
10uF/10V/X5R
DEFDCDC1
EP
DEFDCDC3
AGND1
PWRFAIL_SNS
VCC
VINDCDC2
L2
PGND2
VDCDC2
IC8
TPS650250RHBR
VDCDC3
PGND3
L3
VINDCDC3
VINDCDC1
L1
PGND1
VDCDC1
EN_VDD_ALIVE
MODE
DEFDCDC2
PWRFAIL
EN_DCDC1
EN_DCDC2
EN_DCDC3
EN_LDO
4.7K,5%,1/16W
R65
DEFDCDC1
FB_LDO2
FB_LDO1
VDD_ALIVE
AGND2
VLDO2
VINLDO
VLDO1
+1V8_LDO
+1V2_LDO
Note:
Connect the VDCDCx
Signals near
the load end.
TP2
TP6
TP3
TP7
TP4
TP8
TP5
TestPoint_Alt
5VD_IN
+3V3D
+1V8D
+1V3D
Test Points
TP9
TP10
C73
2.2uF/6.3V/X5R
VDD_ALIVE
4.7K,5%,1/16W
R70
PWRFAIL
Power_Fail Sense
R66
5.6K,1%,1/16W
R71
1.4K,1%,1/16W
PWRFAIL_SNS
Mode Selection
0R,5%,1/10W
R67
0R,5%,1/10W
R72
MODE
Pull-Up
4.7K,5%,1/16W
EN_LDO
R68
EN_DCDC3
R69
AGND Filter
FB3
PGND1
PGND2
PGND3
DEFDCDC2
MMZ2012S121A
AGND2
AGND1
EP
De-Cap
VINLDO
C67
1uF/10V/X5R
VLDO2
C69
2.2uF/6.3V/X5R
VLDO1
C71
2.2uF/6.3V/X5R
LDO-1 Feedback
FB_LDO1
VLDO1
60.4K,1%,1/16W
R56
R61
300K,1%,1/16W
LDO-2 Feedback
200K,1%,1/16W
R62
R63
160K,1%,1/10W
VLDO2
FB_LDO2
Logsun Systems
Sheet: /Power_Supply_II/
File: Power_Supply_II.kicad_sch
Title: DSP320C6748 DVK
Size: A4
Date: 2026-03-28
Rev: 1.0
KiCad E.D.A. 10.0.0
Id: 6/16



Logsun Systems

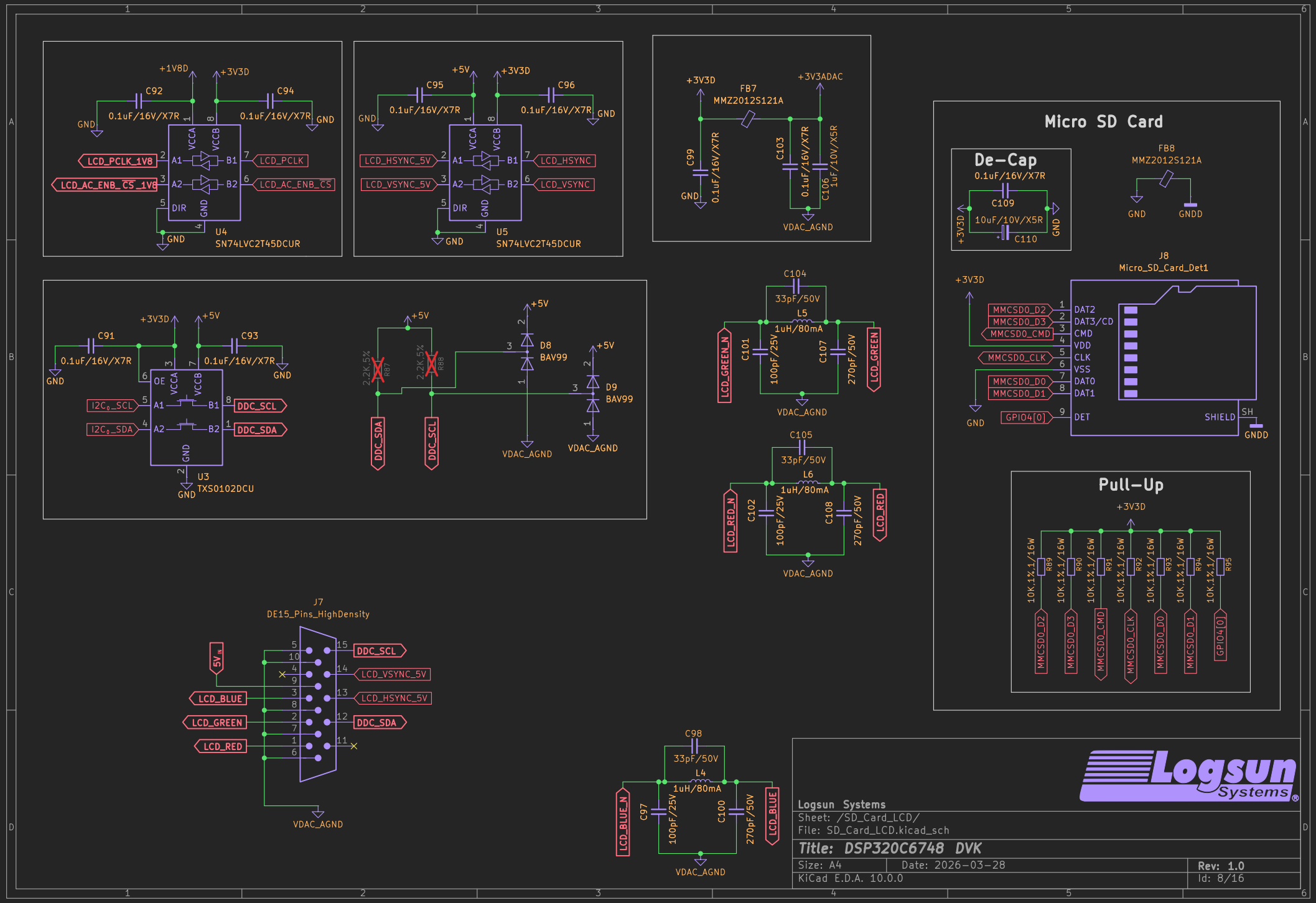
Sheet: /Serial_SATA/

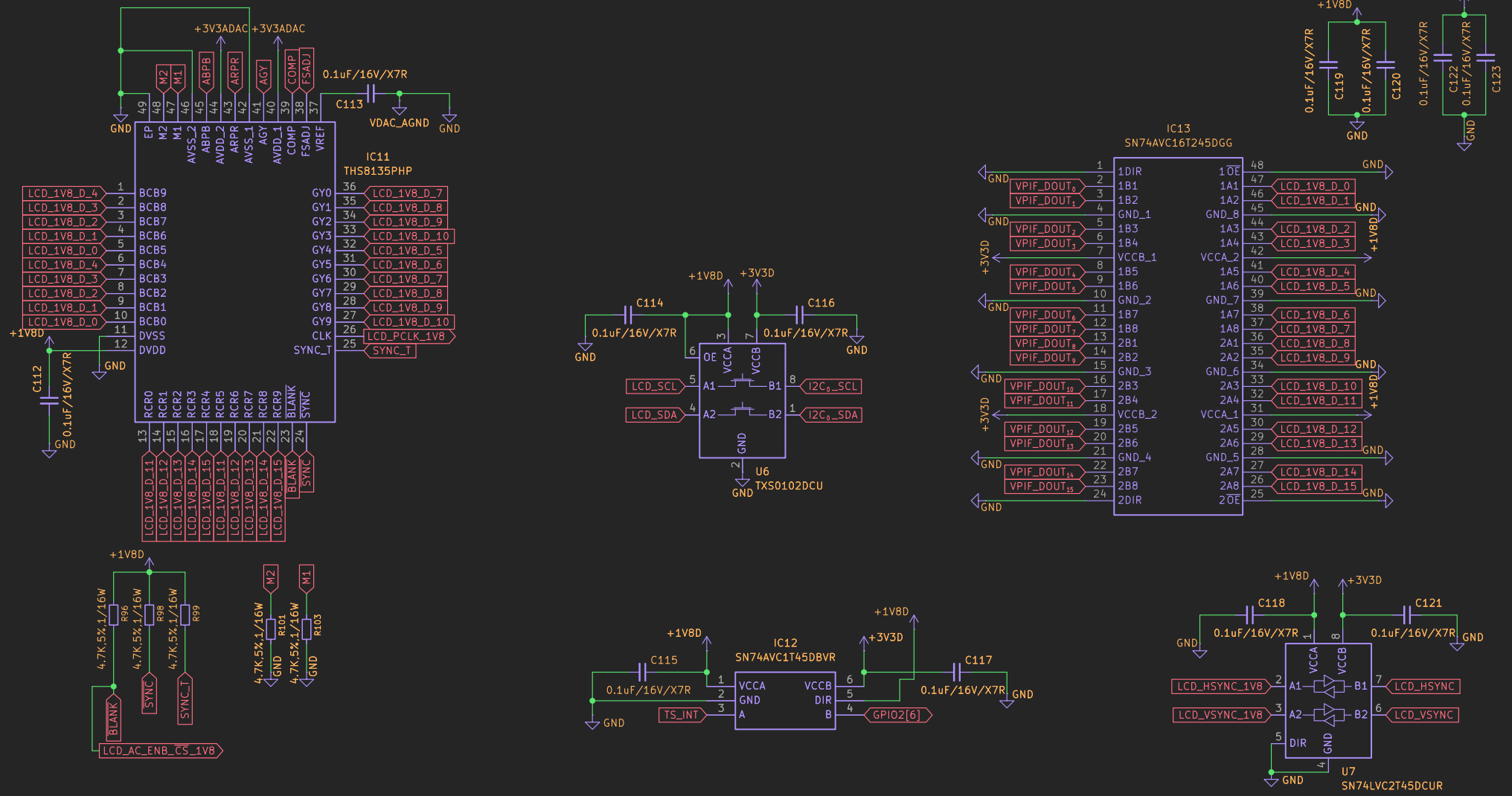
File: Serial_SATA.kicad_sch

Title: DSP320C6748 DVK

Size: A4 | Date: 2026-03-28 | Rev: 1.0

KiCad E.D.A. 10.0.0 | Id: 7/16


+1V8D
+3V3D
C92
0.1uF/16V/X7R
C94
0.1uF/16V/X7R
GND
LCD_PCLK_1V8
LCD_AC_ENB_CS_1V8
VCCA
VCCB
A1
A2
B1
B2
DIR
LCD_PCLK
LCD_AC_ENB_CS
U4
SN74LVC2T45DCUR
+5V
C95
C96
LCD_HSYNC_5V
LCD_VSYNC_5V
LCD_HSYNC
LCD_VSYNC
U5
SN74LVC2T45DCUR
+3V3D
FB7
MMZ2012S121A
+3V3ADAC
C99
0.1uF/16V/X7R
C103
C106
1uF/10V/X5R
VDAC_AGND
C91
C93
OE
I2C0_SCL
I2C0_SDA
DDC_SCL
DDC_SDA
U3
TXS0102DCU
2.2K 5%
R87
R88
D8
BAV99
D9
BAV99
C104
33pF/50V
L5
1uH/80mA
C101
100pF/25V
C107
270pF/50V
LCD_GREEN_N
LCD_GREEN
C105
L6
C102
C108
LCD_RED_N
LCD_RED
J7
DE15_Pins_HighDensity
5V_IN
LCD_BLUE
LCD_GREEN
LCD_RED
LCD_VSYNC_5V
LCD_HSYNC_5V
C98
L4
C97
C100
LCD_BLUE_N
LCD_BLUE
Micro SD Card
De-Cap
C109
10uF/10V/X5R
C110
FB8
MMZ2012S121A
GNDD
J8
Micro_SD_Card_Det1
MMCSD0_D2
MMCSD0_D3
MMCSD0_CMD
MMCSD0_CLK
MMCSD0_D0
MMCSD0_D1
GPIO4[0]
DAT2
DAT3/CD
CMD
VDD
CLK
VSS
DAT0
DAT1
DET
SHIELD
SH
Pull-Up
10K,1%,1/16W
R89
R90
R91
R92
R93
R94
R95
Logsun Systems
Sheet: /SD_Card_LCD/
File: SD_Card_LCD.kicad_sch
Title: DSP320C6748 DVK
Size: A4
Date: 2026-03-28
Rev: 1.0
KiCad E.D.A. 10.0.0
Id: 8/16



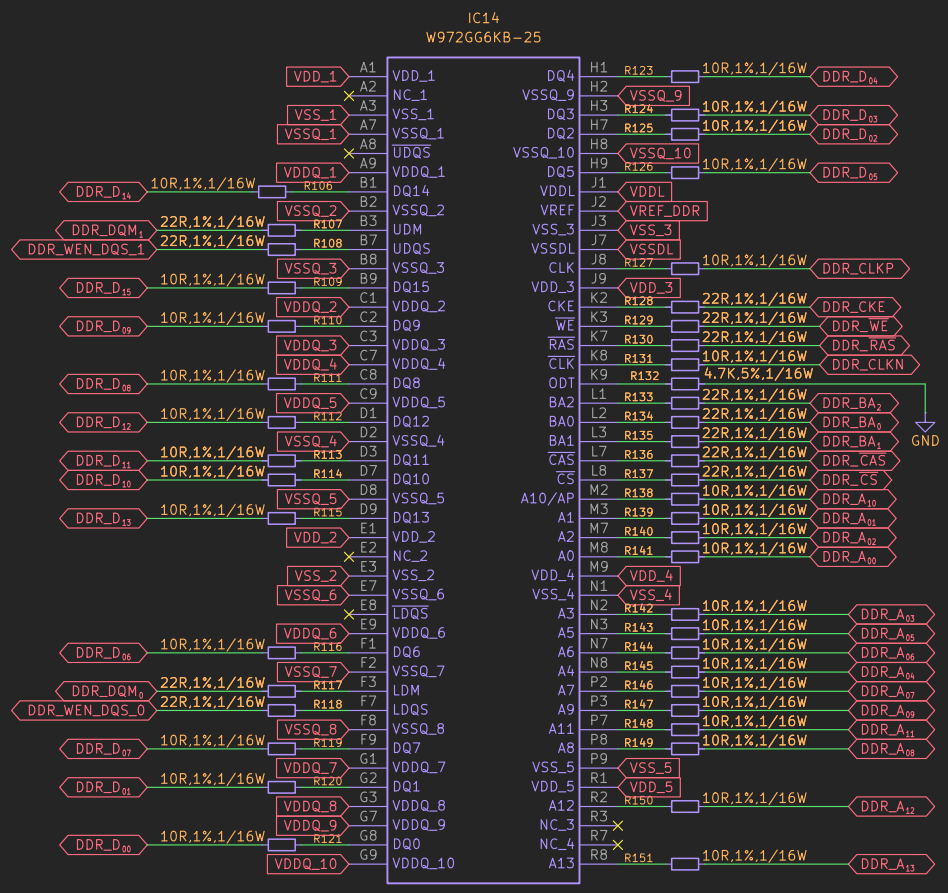
Logsun Systems

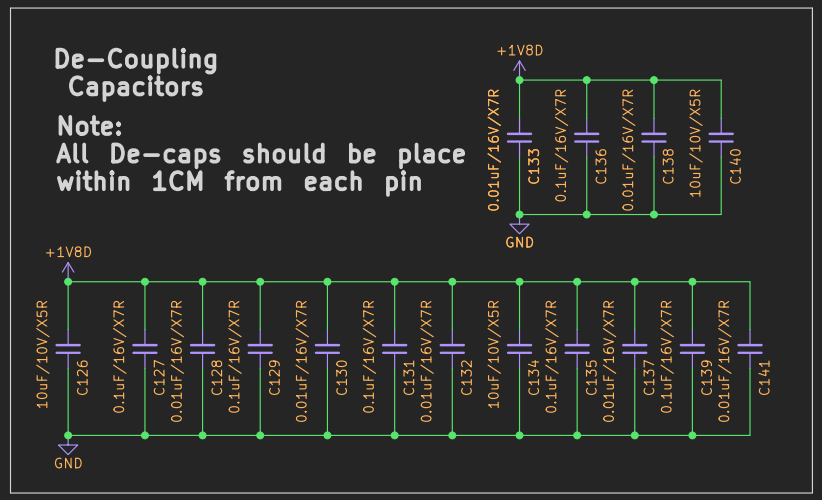
Sheet: /LCD_II/

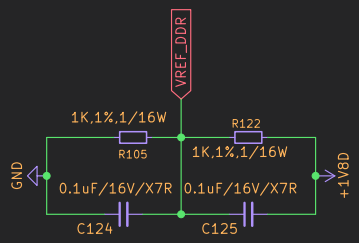
File: LCD_II.kicad_sch

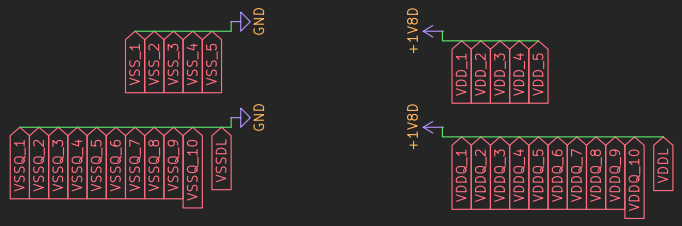
Title: DSP320C6748 DVK

Size: A4	Date: 2026-03-28	Rev: 1.0
KiCad E.D.A. 10.0.0		Id: 9/16







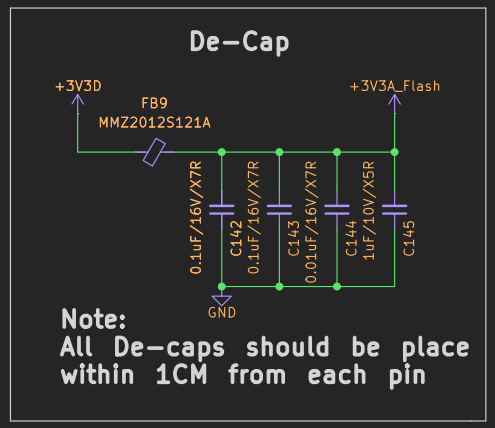


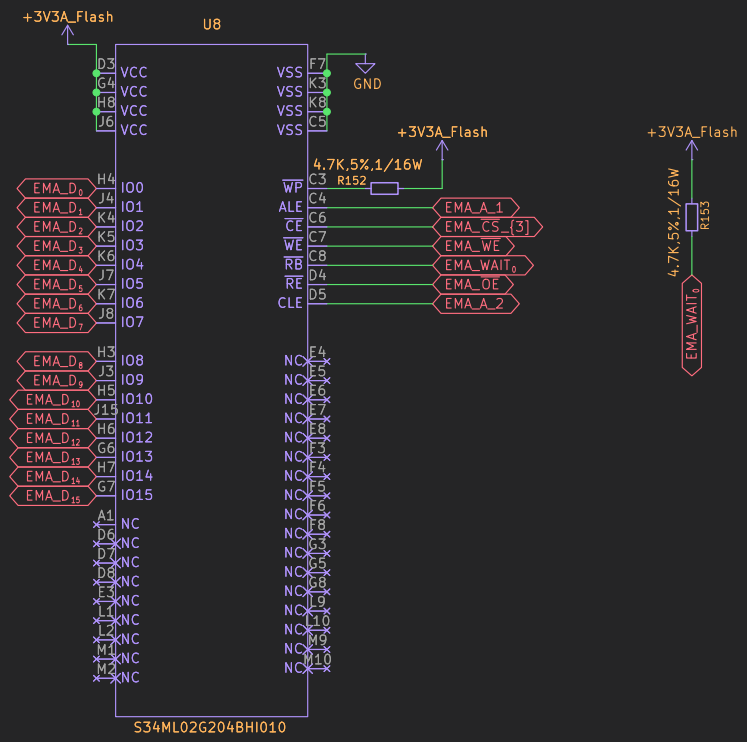
Logsun Systems

Sheet: /DDR_RAM/
File: DDR_RAM.kicad_sch

Title:

Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.0		Id: 10/16



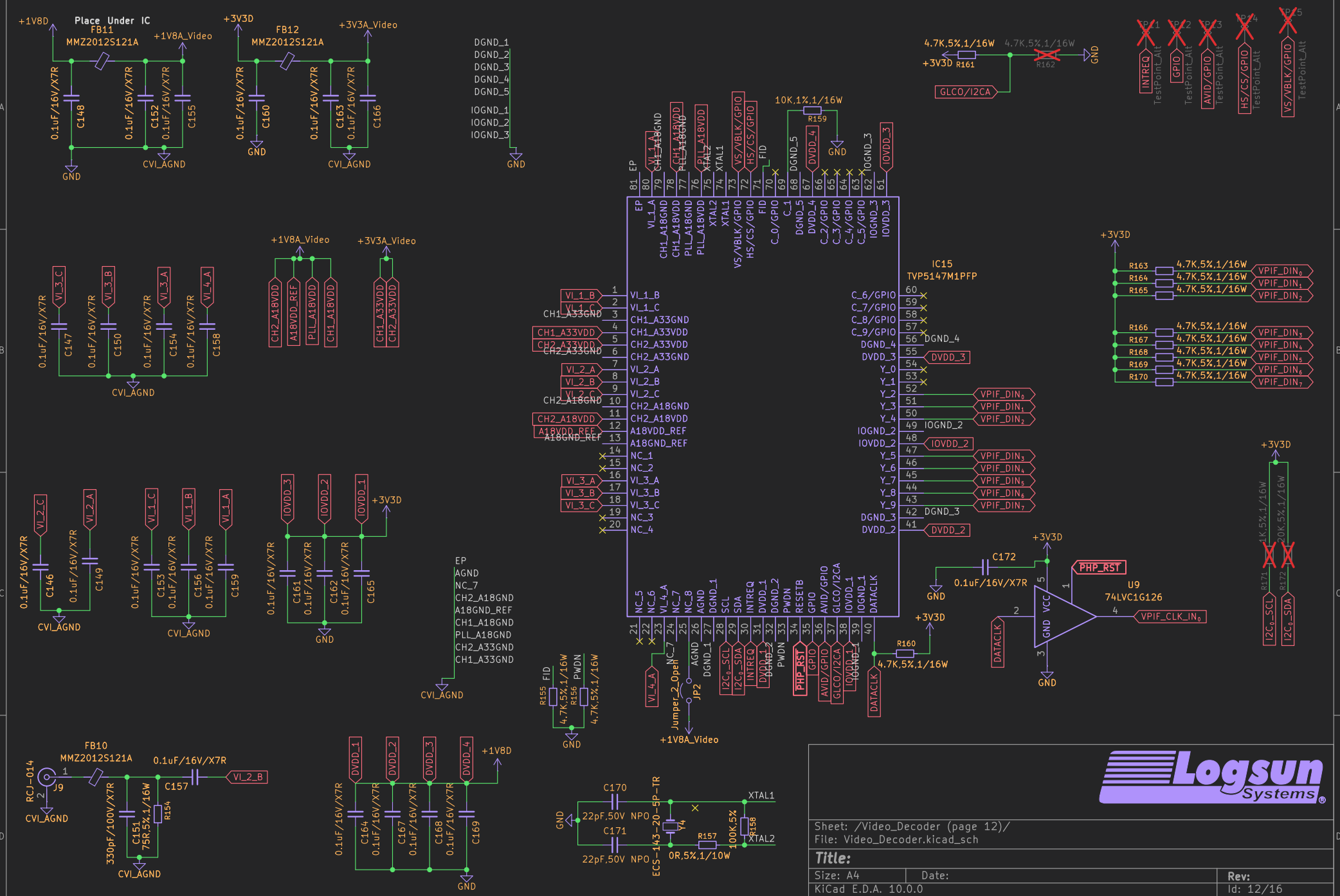


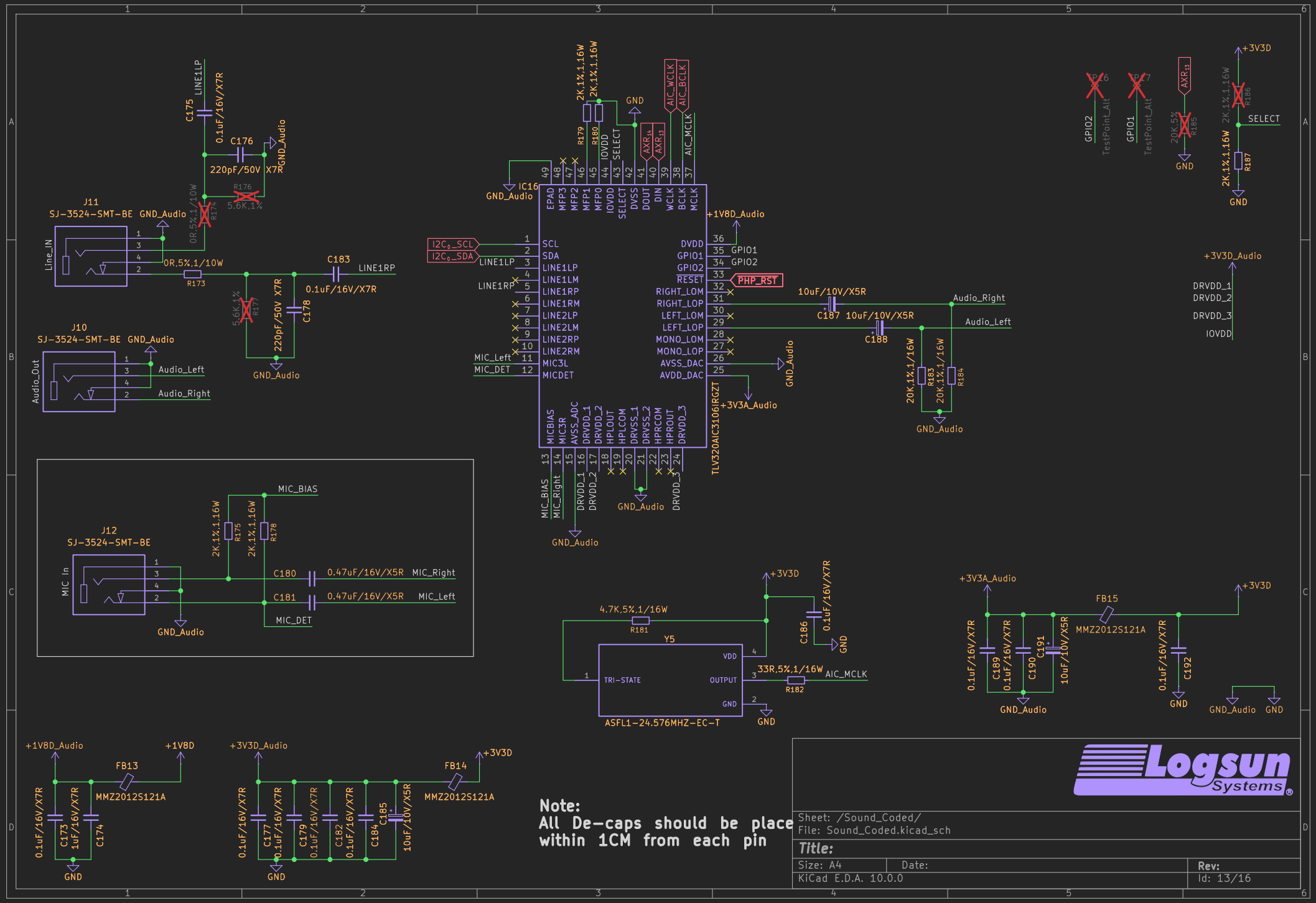
Logsun Systems

Sheet: /NAND_Flash/
File: NAND_Flash.kicad_sch

Title:

Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.0		Id: 11/16


Place Under IC
FB11
MMZ2012S121A
+1V8D
+1V8A_Video
FB12
MMZ2012S121A
+3V3D
+3V3A_Video
0.1uF/16V/X7R
C148
C152
C155
C160
C163
C166
GND
CVI_AGND
DGND_1
DGND_2
DGND_3
DGND_4
DGND_5
IOGND_1
IOGND_2
IOGND_3
4.7K,5%,1/16W
R161
R162
GLCO/I2CA
INTREQ
GPIO
AVID/GPIO
HS/CS/GPIO
VS/VBLK/GPIO
TestPoint_Alt
10K,1%,1/16W
R159
IC15
TVP5147M1PFP
VI_1_A
CH1_A18GND
CH1_A18VDD
PLL_A18GND
PLL_A18VDD
XTAL2
XTAL1
FID
DVDD_4
DVDD_3
IOVDD_3
VI_1_B
VI_1_C
CH1_A33GND
CH1_A33VDD
CH2_A33VDD
CH2_A33GND
VI_2_A
VI_2_B
VI_2_C
CH2_A18GND
CH2_A18VDD
A18VDD_REF
A18GND_REF
NC_1
NC_2
VI_3_A
VI_3_B
VI_3_C
NC_3
NC_4
C_6/GPIO
C_7/GPIO
C_8/GPIO
C_9/GPIO
DGND_4
DVDD_3
Y_0
Y_1
Y_2
Y_3
Y_4
IOGND_2
IOVDD_2
Y_5
Y_6
Y_7
Y_8
Y_9
DGND_3
DVDD_2
VPIF_DIN0
VPIF_DIN1
VPIF_DIN2
VPIF_DIN3
VPIF_DIN4
VPIF_DIN5
VPIF_DIN6
VPIF_DIN7
R163
R164
R165
R166
R167
R168
R169
R170
NC_5
NC_6
NC_7
NC_8
AGND
DGND_1
SCL
SDA
I2C0_SCL
I2C0_SDA
INTREQ
DVDD_1
DGND_2
PWDN
RESETB
GPIO
AVID/GPIO
GLCO/I2CA
IOVDD_1
IOGND_1
DATACLK
PHP_RST
VPIF_CLK_IN0
EP
AGND
NC_7
CH2_A18GND
A18GND_REF
CH1_A18GND
PLL_A18GND
CH2_A33GND
CH1_A33GND
Jumper_2_Open
JP2
R155
R156
4.7K,5%,1/16W
R160
C172
0.1uF/16V/X7R
U9
74LVC1G126
VCC
DATACLK
R171
R172
1K,5%,1/16W
20K,5%,1/16W
FB10
MMZ2012S121A
RCJ-014
J9
C157
VI_2_B
330pF/100V/X7R
C151
75R,5%,1/16W
R154
C164
C167
C168
C169
C170
C171
22pF,50V NPO
ECS-143-20-5P-TR
Y4
R157
0R,5%,1/10W
100K,5%
R158
XTAL1
XTAL2
Logsun Systems
Sheet: /Video_Decoder (page 12)/
File: Video_Decoder.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.0
Id: 12/16


J11
SJ-3524-SMT-BE
GND_Audio
Line_IN
C175
LINE1LP
0.1uF/16V/X7R
C176
220pF/50V X7R
GND_Audio
R176
5.6K,1%
R174
0R,5%,1/10W
0R,5%,1/10W
R173
C183
LINE1RP
0.1uF/16V/X7R
5.6K,1%
R177
220pF/50V X7R
C178
GND_Audio
J10
SJ-3524-SMT-BE
GND_Audio
Audio_Out
Audio_Left
Audio_Right
IC16
GND_Audio
2K,1%,1.16W
2K,1%,1.16W
R179
R180
IOVDD
SELECT
GND
AIC_WCLK
AIC_BCLK
AXR14
AXR13
AIC_MCLK
EPAD
MFP3
MFP2
MFP1
MFP0
IOVDD
SELECT
DVSS
DOUT
DIN
WCLK
BCLK
MCLK
I2C0_SCL
I2C0_SDA
LINE1LP
LINE1RP
MIC_Left
MIC_DET
SCL
SDA
LINE1LP
LINE1LM
LINE1RP
LINE1RM
LINE2LP
LINE2LM
LINE2RP
LINE2RM
MIC3L
MICDET
+1V8D_Audio
DVDD
GPIO1
GPIO2
RESET
PHP_RST
RIGHT_LOM
RIGHT_LOP
LEFT_LOM
LEFT_LOP
MONO_LOM
MONO_LOP
AVSS_DAC
AVDD_DAC
GND_Audio
+3V3A_Audio
TLV320AIC3106IRGZT
MICBIAS
MIC3R
AVSS_ADC
DRVDD_1
DRVDD_2
HPLOUT
HPLCOM
DRVSS_1
DRVSS_2
HPRCOM
HPROUT
DRVDD_3
MIC_BIAS
MIC_Right
DRVDD_1
DRVDD_2
GND_Audio
GND_Audio
DRVDD_3
10uF/10V/X5R
C187
10uF/10V/X5R
C188
Audio_Right
Audio_Left
20K,1%,1/16W
R183
20K,1%,1/16W
R184
GND_Audio
GPIO2
TestPoint_Alt
GPIO1
TestPoint_Alt
AXR13
20K,5%
R185
GND
+3V3D
2K,1%,1.16W
R186
SELECT
2K,1%,1.16W
R187
GND
+3V3D_Audio
DRVDD_1
DRVDD_2
DRVDD_3
IOVDD
J12
SJ-3524-SMT-BE
MIC In
GND_Audio
2K,1%,1.16W
R175
2K,1%,1.16W
R178
MIC_BIAS
C180
0.47uF/16V/X5R
MIC_Right
C181
0.47uF/16V/X5R
MIC_Left
MIC_DET
+3V3D
4.7K,5%,1/16W
R181
C186
0.1uF/16V/X7R
GND
Y5
VDD
TRI-STATE
OUTPUT
GND
33R,5%,1/16W
R182
AIC_MCLK
GND
ASFL1-24.576MHZ-EC-T
+3V3A_Audio
FB15
MMZ2012S121A
+3V3D
0.1uF/16V/X7R
C189
0.1uF/16V/X7R
C190
C191
10uF/10V/X5R
0.1uF/16V/X7R
C192
GND_Audio
GND
GND_Audio
GND
+1V8D_Audio
FB13
MMZ2012S121A
+1V8D
0.1uF/16V/X7R
C173
1uF/16V/X7R
C174
GND
+3V3D_Audio
FB14
MMZ2012S121A
+3V3D
0.1uF/16V/X7R
C177
0.1uF/16V/X7R
C179
0.1uF/16V/X7R
C182
0.1uF/16V/X7R
C184
C185
10uF/10V/X5R
GND
Note:
All De-caps should be place
within 1CM from each pin
Logsun Systems
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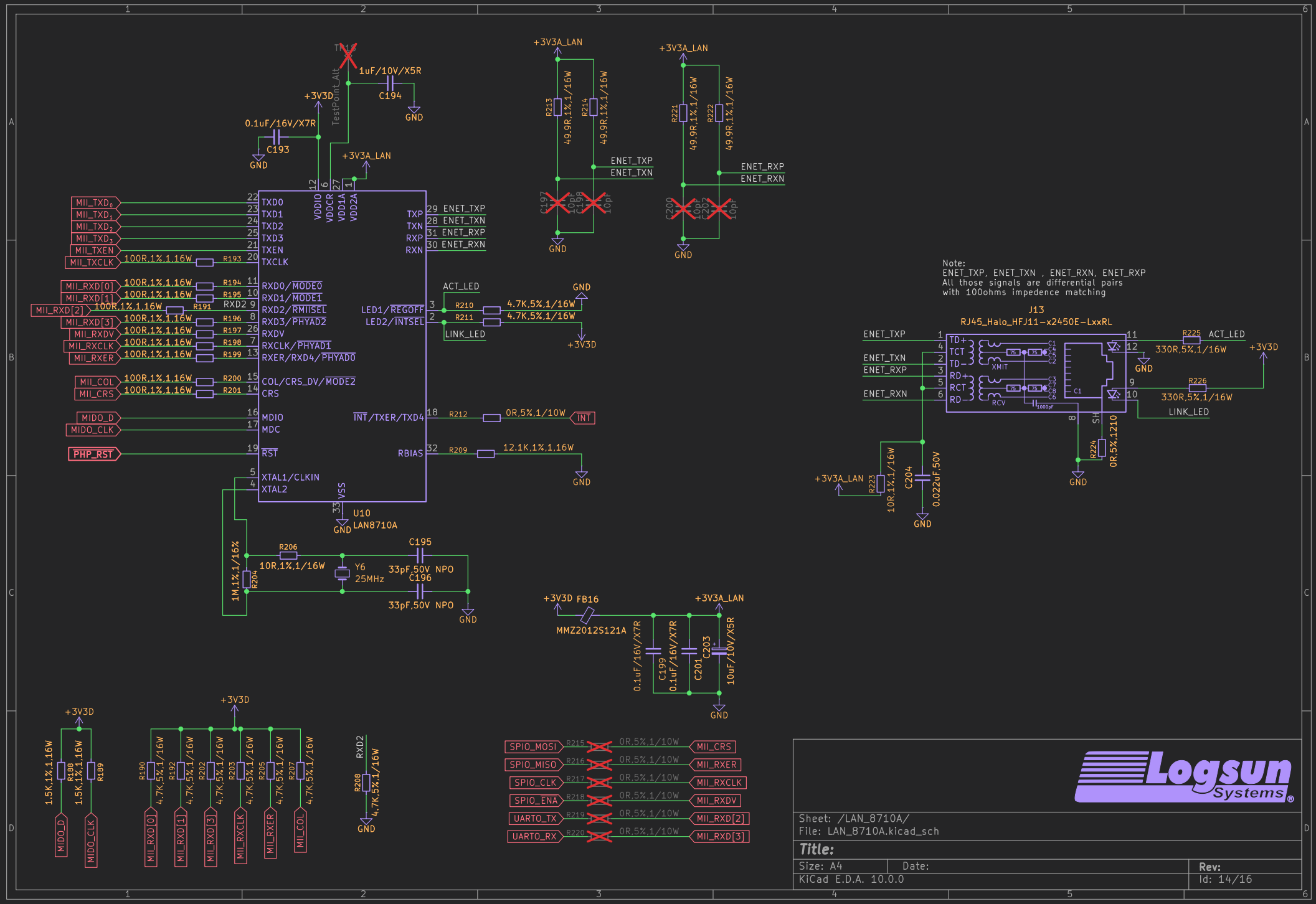
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.0
Id: 13/16


U10
LAN8710A
TXD0
TXD1
TXD2
TXD3
TXEN
TXCLK
RXD0/MODE0
RXD1/MODE1
RXD2/RMIISEL
RXD3/PHYAD2
RXDV
RXCLK/PHYAD1
RXER/RXD4/PHYAD0
COL/CRS_DV/MODE2
CRS
MDIO
MDC
RST
XTAL1/CLKIN
XTAL2
VDDIO
VDDCR
VDD1A
VDD2A
VSS
TXP
TXN
RXP
RXN
LED1/REGOFF
LED2/INTSEL
INT/TXER/TXD4
RBIAS
ENET_TXP
ENET_TXN
ENET_RXP
ENET_RXN
ACT_LED
LINK_LED
INT
MII_TXD0
MII_TXD1
MII_TXD2
MII_TXD3
MII_TXEN
MII_TXCLK
MII_RXD[0]
MII_RXD[1]
MII_RXD[2]
MII_RXD[3]
MII_RXDV
MII_RXCLK
MII_RXER
MII_COL
MII_CRS
MIDO_D
MIDO_CLK
PHP_RST
RXD2
R193 100R,1%,1,16W
R194 100R,1%,1,16W
R195 100R,1%,1,16W
R191 100R,1%,1,16W
R196 100R,1%,1,16W
R197 100R,1%,1,16W
R198 100R,1%,1,16W
R199 100R,1%,1,16W
R200 100R,1%,1,16W
R201 100R,1%,1,16W
R210 4.7K,5%,1/16W
R211 4.7K,5%,1/16W
R212 0R,5%,1/10W
R209 12.1K,1%,1,16W
R206 10R,1%,1/16W
R204 1M,1%,1/16%
Y6 25MHz
C195 33pF,50V NPO
C196 33pF,50V NPO
C194 1uF/10V/X5R
C193 0.1uF/16V/X7R
TP18 TestPoint_Alt
+3V3D
+3V3A_LAN
GND
R213 49.9R,1%,1/16W
R214 49.9R,1%,1/16W
R221 49.9R,1%,1/16W
R222 49.9R,1%,1/16W
C197 10pF
C198 10pF
C200 10pF
C202 10pF
Note:
ENET_TXP, ENET_TXN , ENET_RXN, ENET_RXP
All those signals are differential pairs
with 100ohms impedence matching
J13
RJ45_Halo_HFJ11-x2450E-LxxRL
TD+
TCT
TD-
RD+
RCT
RD-
XMIT
RCV
R225 330R,5%,1/16W
R226 330R,5%,1/16W
R224 0R,5%,1210
R223 10R,1%,1/16W
C204 0.022uF,50V
FB16 MMZ2012S121A
C199 0.1uF/16V/X7R
C201 0.1uF/16V/X7R
C203 10uF/10V/X5R
R188 1.5K,1%,1,16W
R189 1.5K,1%,1,16W
R190 4.7K,5%,1/16W
R192 4.7K,5%,1/16W
R202 4.7K,5%,1/16W
R203 4.7K,5%,1/16W
R205 4.7K,5%,1/16W
R207 4.7K,5%,1/16W
R208 4.7K,5%,1/16W
SPIO_MOSI R215 0R,5%,1/10W MII_CRS
SPIO_MISO R216 0R,5%,1/10W MII_RXER
SPIO_CLK R217 0R,5%,1/10W MII_RXCLK
SPIO_ENA R218 0R,5%,1/10W MII_RXDV
UART0_TX R219 0R,5%,1/10W MII_RXD[2]
UART0_RX R220 0R,5%,1/10W MII_RXD[3]
Logsun Systems
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Size: A4
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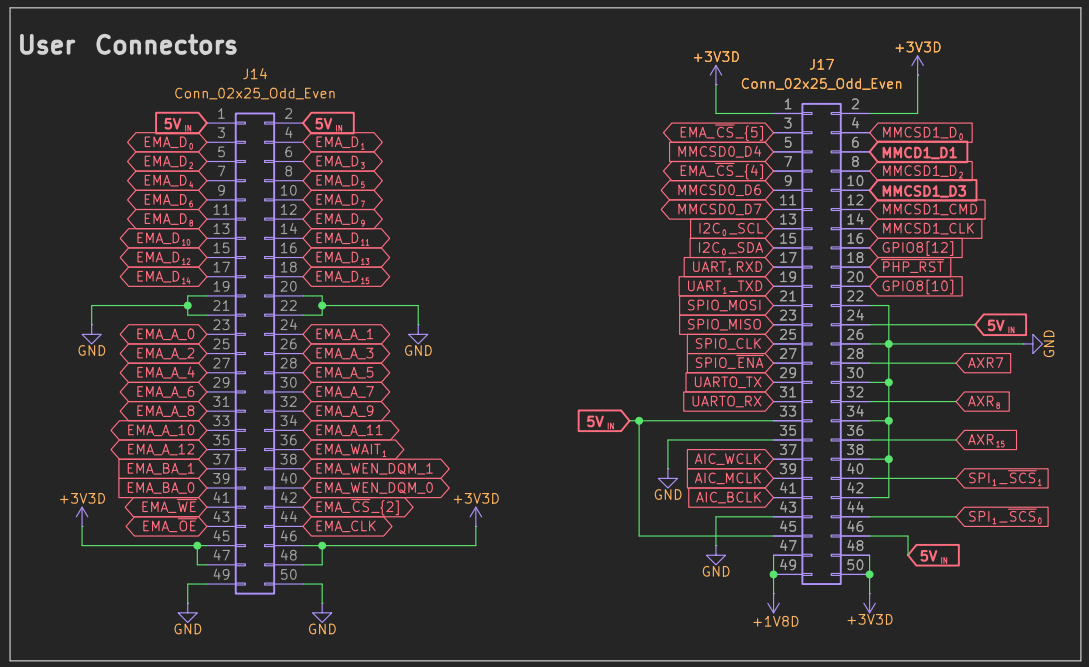

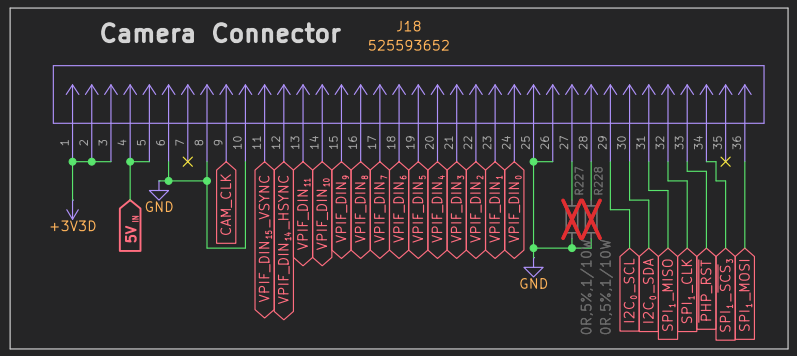

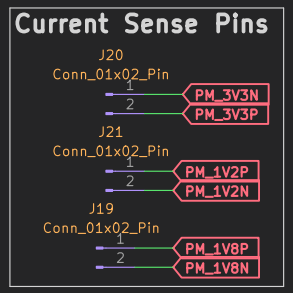

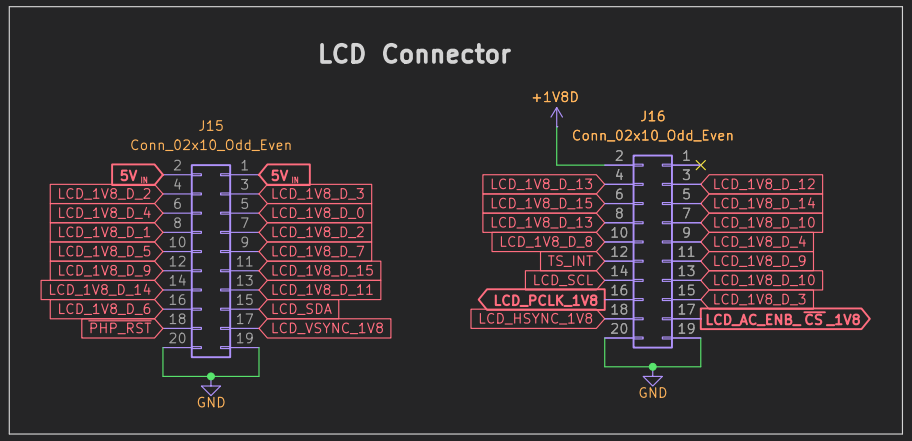

GPIO0[8]

Logsun Systems

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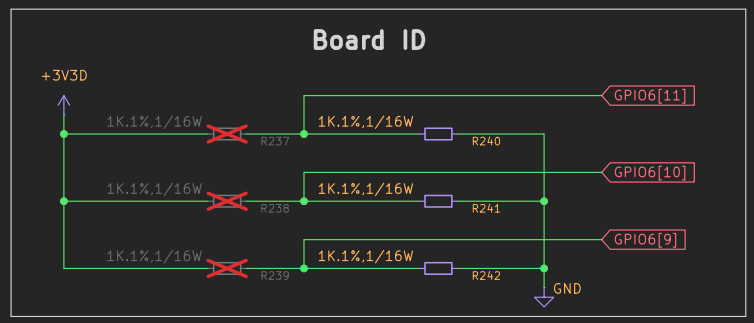

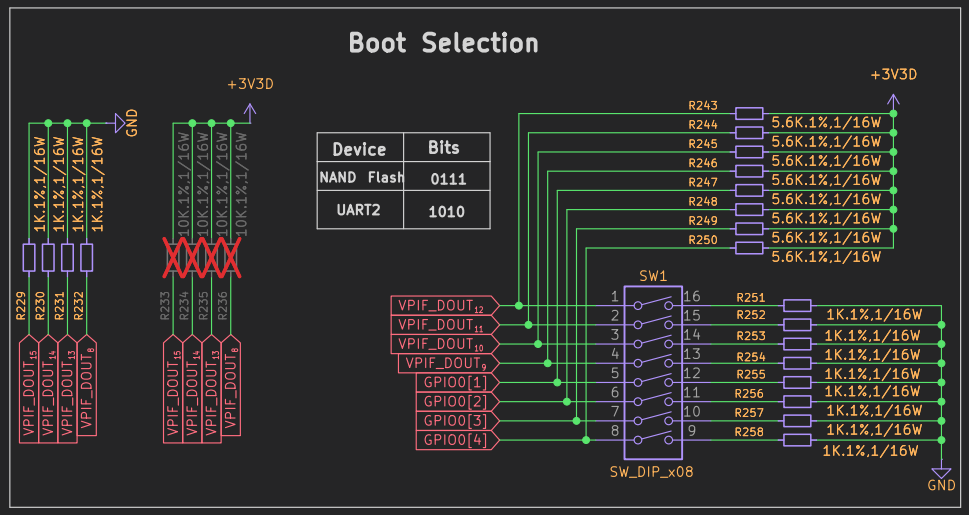

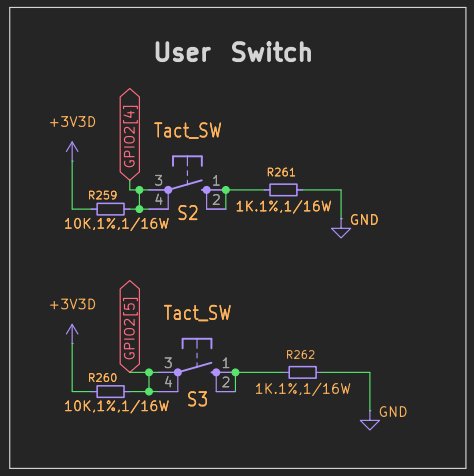

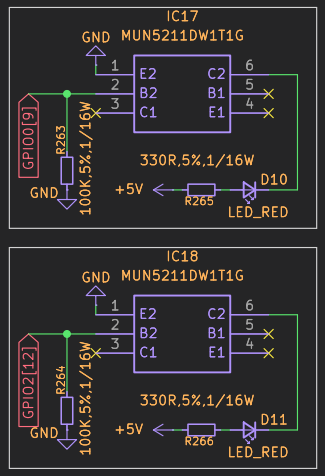

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